Simulating Spin Dephasing from Measured Magnetic Field Noise: Ramsey, Hahn Echo and CPMG Dynamical Decoupling*

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Abstract

We simulate the dephasing of an ensemble of one hundred spin qubits driven by experimentally recorded traces of effective magnetic field noise. Each trace is treated as the noise realisation experienced by one spin, and the qubit state is propagated as a density matrix through explicit instantaneous rotation operators and free evolution governed by the accumulated stochastic phase. The noise power spectral density is estimated by two independent routes, Welch periodogram averaging over the ensemble and the Fourier transform of the ensemble averaged autocorrelation function, and is found to follow an inverse square frequency law across the accessible band, consistent with a Lorentzian spectrum whose knee lies below the inverse record length. Coherence decay curves are computed for a free induction Ramsey sequence, a Hahn echo, and Carr Purcell Meiboom Gill sequences with two, four and eight refocusing pulses. Fitting a stretched exponential to each curve yields a free induction dephasing time of seventy one microseconds with a near Gaussian exponent, rising to four point four milliseconds under eight pulse decoupling. Coherence times predicted from the measured spectrum through the filter function formalism agree with the trace based simulation to within ten percent with no adjustable parameters, and the extracted decoupling gain follows a two thirds power law in pulse number as expected for Lorentzian noise.

Index Terms— spin qubit, decoherence, dephasing, density matrix, dynamical decoupling, Hahn echo, CPMG, power spectral density, filter function

I Introduction

The practical utility of a spin qubit is bounded by the time over which it retains a well defined quantum phase. Isolated quantum systems evolve unitarily and reversibly, but a real spin is embedded in an environment whose fluctuating degrees of freedom imprint a random phase on the qubit. When these fluctuations couple to the spin as an effective magnetic field along the quantisation axis, the resulting loss of phase information is called dephasing, and it is typically the dominant error channel in solid state spin qubits.

Two questions follow. First, given a measured record of the environmental noise, can one predict quantitatively how fast coherence will decay? Second, how much of that decay can be recovered by pulse sequences that refocus the accumulated phase? This paper addresses both using an ensemble of one hundred recorded noise traces. Section II develops the theory: unitary evolution under the Schrödinger equation, the density matrix description of a mixed ensemble, and the pulse sequences used to suppress dephasing. Section III describes the estimation of the noise power spectral density and the filter function formalism that links it to coherence decay. Section IV sets out the numerical implementation, Section V presents the simulated decay curves and extracted coherence times, and Section VI interprets the results.

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II Theory

A Unitary evolution

The state $|\psi(t)\rangle$ of a closed quantum system obeys the time dependent Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle , \tag{1}$$

where $\hat{H}$ is the Hamiltonian. For a time independent Hamiltonian the solution is generated by the unitary propagator

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle , \qquad \hat{U}(t) = e^{-i\hat{H}t/\hbar}. \tag{2}$$

For a spin one half in a magnetic field $\mathbf{B}$ the interaction is the Zeeman coupling $\hat{H} = -\gamma \mathbf{B} \cdot \hat{\mathbf{S}}$, with γ the gyromagnetic ratio and $\hat{\mathbf{S}} = (\hbar/2)\boldsymbol{\sigma}$ the spin operator. A field along z therefore produces the Hamiltonian $\hat{H}_0 = -\gamma B_z \hat{S}_z$ and a precession of the transverse spin components at the Larmor frequency $\omega_0 = \gamma B_z$.

Equation (2) describes perfect, reversible evolution. It cannot by itself describe decoherence, because a unitary map preserves the purity $\mathrm{Tr}(\rho^2)$ of a state. Decoherence arises when the system is coupled to an environment whose state is not tracked.

B Density matrix and coherence

An ensemble of spins in states $|\psi_i\rangle$ with probabilities p_i is described by the density matrix

$$\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i| , \tag{3}$$

which evolves under the von Neumann equation $\dot{\rho} = -\frac{i}{\hbar}[\hat{H}, \rho]$ for unitary dynamics, and under the Lindblad master equation

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} \left[\hat{H}, \rho\right] + \mathcal{L}(\rho) \tag{4}$$

when the environment is traced out, the superoperator $\mathcal{L}$ carrying the dissipative and dephasing terms. In the basis

$\{|0\rangle, |1\rangle\}$ the diagonal elements ρ_{00} and ρ_{11} are the populations, while the off diagonal element ρ_{01} is the coherence, encoding the definiteness of the relative phase. Pure dephasing is the decay of ρ_{01} at fixed populations.

The present work does not postulate a Lindblad operator. Instead, since the noise record is available explicitly, each spin is evolved unitarily under its own realisation of the field and the ensemble average is taken at the end. This is exact for classical noise and does not require the Markov approximation implicit in (4); the irreversible looking decay emerges from averaging over realisations rather than being imposed.

C Stochastic phase accumulation

Let spin i experience the fluctuating field $B_i(t)$ along z, so that

$$\hat{H}_i(t) = -\gamma B_i(t)\, \hat{S}_z = -\frac{\hbar}{2}\,\gamma B_i(t)\, \sigma_z. \tag{5}$$

Because $\hat{H}_i(t)$ commutes with itself at different times, the propagator is obtained without time ordering,

$$\hat{U}_i(t) = \exp\left[-\frac{i}{2}\phi_i(t)\,\sigma_z\right], \quad \phi_i(t) = \gamma\int_0^t B_i(t')\,dt'. \tag{6}$$

The random variable ϕ_i is the accumulated phase. Starting from the superposition $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, the coherence of spin i acquires the factor $e^{-i\phi_i}$, and the ensemble averaged coherence is

$$W(\tau) \equiv 2\,|\langle\rho_{01}\rangle| = \left|\left\langle e^{-i\phi}\right\rangle\right|. \tag{7}$$

It is worth being precise about what decays here. Because the noise is classical and couples only along z, the state of spin i remains exactly pure, $|\psi_i\rangle = (|0\rangle + e^{-i\phi_i}\,|1\rangle)/\sqrt{2}$ with $\mathrm{Tr}(\rho_i^2) = 1$ at all times: its Bloch vector retains unit length and merely precesses to a different azimuthal angle. The populations likewise remain fixed at $\rho_{00} = \rho_{11} = 1/2$, since pure dephasing transfers no population. What decays in (7) is the resultant of an ensemble of unit vectors fanning out across the equator, which is a property of the average rather than of any individual spin.

This is nonetheless what is observed experimentally. The concluding $\pi/2$ pulse maps the accumulated phase onto a population difference, $P(|0\rangle) = \cos^2(\phi_i/2)$, so a single shot returns 0 or 1 at random through quantum projection noise, while the bias of that outcome varies from shot to shot as ϕ_i varies. Averaging recovers $\langle P(|0\rangle)\rangle = [1+W(\tau)]/2$, so the spread of ϕ appears directly as a loss of readout contrast.

The distinction between a fanned out ensemble of pure states and genuine decoherence is precisely what the refocusing sequences of Section D exploit. Each phase is intact but unknown to the observer, so a π pulse can reverse its accumulation and reconverge the fan; had the spins entangled with a quantum bath, no pulse sequence could recover the lost information. The ensemble averaged $\langle\rho\rangle$ is genuinely mixed in either case, and the two situations are indistinguishable from ensemble averaged measurements alone, differing only in their response to refocusing. It should be emphasised that the purity of the individual states is a consequence of the classical, purely longitudinal noise assumed here, and is not a general property of spin qubits.

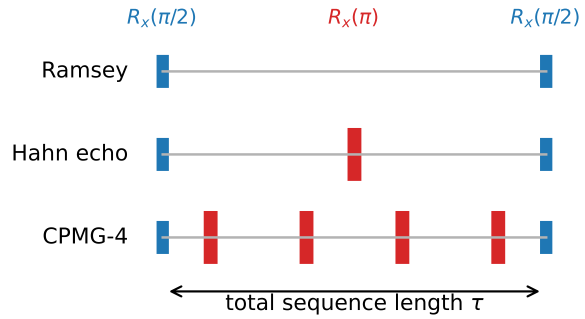

Figure 1: Pulse sequences studied. Narrow bars are $\pi/2$ pulses about x, wide bars are π refocusing pulses. All pulses are instantaneous and the abscissa is the total sequence length τ.

For zero mean Gaussian noise the average can be evaluated in closed form,

$$W(\tau) = \exp\left[-\tfrac{1}{2}\left\langle\phi^2(\tau)\right\rangle\right] \equiv e^{-\chi(\tau)}, \tag{8}$$

which defines the decoherence function χ.

Two limits are instructive. If the noise is effectively static over the measurement, $\phi = \gamma B\tau$ with B drawn from a distribution of width σ_B, giving a Gaussian decay

$$W(\tau) = \exp\left[-\tfrac{1}{2}\gamma^2\sigma_B^2\tau^2\right], \qquad T_2^* = \frac{\sqrt{2}}{\gamma\sigma_B}, \tag{9}$$

where T_2^* is defined by $W(T_2^*) = e^{-1}$. If instead the noise is delta correlated, $\langle\phi^2\rangle$ grows linearly in τ and the decay is a simple exponential. The exponent p in the stretched exponential $W = \exp[-(\tau/T_2)^p]$ therefore diagnoses the noise correlation time.

D Pulse sequences

All sequences considered here begin by rotating the ground state $|0\rangle$ onto the equator of the Bloch sphere with the instantaneous rotation

$$R_x(\theta) = \exp\left[-\frac{i\theta}{2}\sigma_x\right] = \cos\frac{\theta}{2}\mathbb{1} - i\sin\frac{\theta}{2}\,\sigma_x, \tag{10}$$

taking $\theta = \pi/2$. They differ in the refocusing pulses applied during the subsequent evolution, as sketched in Fig. 1.

Ramsey. The spin evolves freely for the whole interval τ and the phase (6) accumulates without interruption. All noise frequencies down to zero contribute, so the static spread of the field is fully sampled and the decay time is the inhomogeneous dephasing time T_2^*. A second $\pi/2$ pulse converts the surviving coherence into a population difference for readout.

Hahn echo. A single π pulse at $\tau/2$ inverts the spin, so the phase accumulated in the second half enters with the opposite sign. The net phase becomes

$$\phi = \gamma\left[\int_0^{\tau/2} B\,dt' - \int_{\tau/2}^{\tau} B\,dt'\right], \tag{11}$$

which vanishes identically for a static field. The echo therefore removes the quasi static contribution and the resulting decay time is the intrinsic T_2.

CPMG. Extending the idea, N refocusing pulses are placed at

$$t_j = \frac{(2j-1)\tau}{2N}, \qquad j = 1, \ldots, N, \tag{12}$$

the Carr Purcell Meiboom Gill timing; the Hahn echo is the case $N = 1$. Each additional pulse shortens the interval over which phase accrues coherently, so the sequence becomes progressively blind to low frequency noise. Because the pulses are applied about the same axis as the initial rotation, CPMG is also robust against small pulse amplitude errors, although this is immaterial here since all pulses are taken to be ideal and instantaneous.

III Noise Spectrum and Filter Functions

A Spectral estimation

The noise is assumed stationary and ergodic, so ensemble and time averages coincide and the autocorrelation function

$$R_B(s) = \langle B(t)B(t+s)\rangle \tag{13}$$

depends only on the lag s. The Wiener Khinchin theorem relates it to the one sided power spectral density,

$$S_B(f) = 2\int_{-\infty}^{\infty} R_B(s)\, e^{-i2\pi f s}\, ds, \qquad f \geq 0, \tag{14}$$

normalised so that $\sigma_B^2 = \int_0^\infty S_B(f)\, df$.

Two estimators are used. The first evaluates (14) directly: the autocorrelation of each trace is computed by fast Fourier transform, averaged over the ensemble, tapered with a Hann lag window to suppress the poorly determined long lag estimates, and transformed. The second is Welch's method, applied to each trace over its full length with a Hann data window and then averaged across the one hundred traces; ergodicity makes this ensemble average a legitimate substitute for the segment averaging normally used to reduce periodogram variance, at no cost in frequency resolution.

B From spectrum to coherence

For Gaussian noise the decoherence function in (8) can be written as an overlap between the noise spectrum and a filter determined entirely by the pulse timing,

$$\chi(\tau) = \frac{1}{2\pi}\int_0^\infty S_\omega(\omega)\left|\tilde{f}(\omega,\tau)\right|^2 d\omega, \tag{15}$$

where $S_\omega(\omega) = \gamma^2 S_B(\omega/2\pi)/4\pi$ is the angular frequency spectrum of the phase noise and

$$\tilde{f}(\omega,\tau) = \int_0^\tau y(t)\, e^{i\omega t}\, dt \tag{16}$$

is the Fourier transform of the toggling function $y(t) = \pm 1$, which switches sign at each π pulse. Equation (15) makes the role of the sequences transparent: Ramsey has $y = +1$ throughout, so $|\tilde{f}|^2$ peaks at $\omega = 0$ and the sequence is maximally sensitive to slow noise; each refocusing pulse pushes the passband up to $\omega \approx \pi N/\tau$, where a decreasing spectrum supplies less power. Because $\tilde{f}$ is piecewise analytic, the integral in (16) reduces to a sum over pulse edges and is evaluated without further approximation.

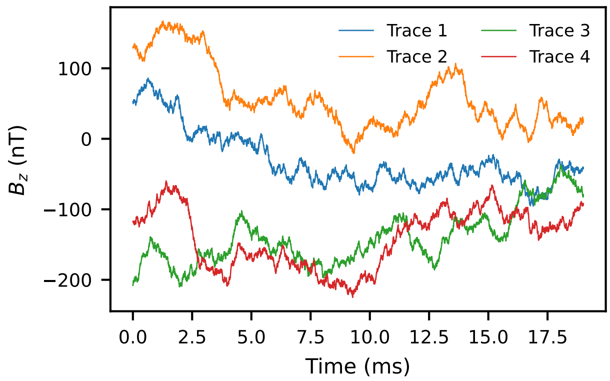

Figure 2: Four of the one hundred recorded traces of effective magnetic field noise. The traces wander on millisecond timescales and carry substantial static offsets, both signatures of a strongly low frequency weighted spectrum.

IV Numerical Method

The data consist of one hundred traces of effective field noise sampled at $\Delta t = 2.5$ μs for 7603 points, a record length of 19.01 ms per trace. The gyromagnetic ratio is taken as $\gamma = 2\pi \times 28$ GHz T^{-1}, appropriate for an electron spin.

Each trace is treated as the field seen by one spin. The phase integral in (6) is evaluated as a cumulative sum,

$$\phi_i(t_k) = \gamma\,\Delta t \sum_{j<k} B_i(t_j), \tag{17}$$

which amounts to holding the field constant across each sample interval. Since the traces vary on millisecond scales, far slower than Δt, the associated error is negligible. Storing the cumulative phase allows the phase accrued in any interval between pulses to be obtained by a single subtraction.

For a given sequence and total length τ, the density matrix of every spin is initialised as $\rho = |0\rangle\langle 0|$, rotated by $R_x(\pi/2)$, and then propagated segment by segment. Within each free evolution segment the map is

$$\rho \to U(\phi)\,\rho\, U^\dagger(\phi), \qquad U(\phi) = \mathrm{diag}\left(e^{-i\phi/2},\, e^{+i\phi/2}\right), \tag{18}$$

and between segments the refocusing operation $\rho \to R_x(\pi)\,\rho\, R_x^\dagger(\pi)$ is applied explicitly rather than absorbed into a sign convention. The one hundred density matrices are then averaged and the coherence read off via (7). Sweeping τ over the sampling grid produces the decay curve. Pulse positions are rounded to the nearest sample, an error bounded by Δt and negligible on the timescales of interest.

Each decay curve is fitted with the stretched exponential

$$W(\tau) = \exp\left[-(\tau/T_2)^p\right], \tag{19}$$

with T_2 and p free. Only points above $W = 0.15$ are fitted, since a finite ensemble of N_{tr} traces cannot resolve coherence below the statistical floor $\sqrt{\pi}/2\sqrt{N_{\mathrm{tr}}} \approx 0.09$.

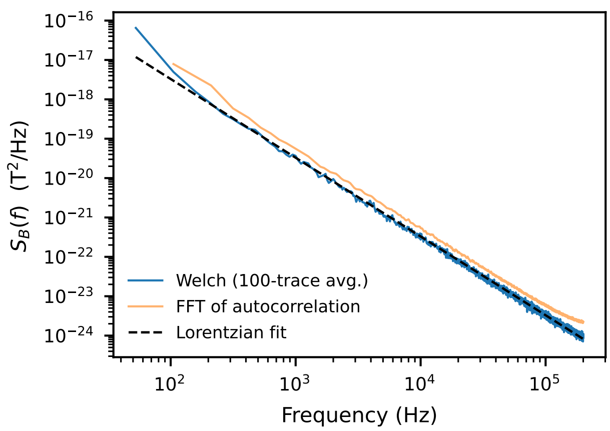

Figure 3: Estimated one sided power spectral density of the field noise from Welch periodogram averaging over the ensemble and from the Fourier transform of the averaged autocorrelation function, with a Lorentzian fit. The spectrum follows f^{-2} over the whole band; the constant offset of the direct estimator is window leakage, as discussed in the text.

V Results

A Noise characterisation

Representative traces are shown in Fig. 2. The ensemble standard deviation is $\sigma_B = 95.1$ nT with a residual mean offset of 2.5 nT. The traces are visibly correlated over their whole length and differ markedly in their mean value, which is the hallmark of quasi static inhomogeneous broadening.

The estimated spectrum is shown in Fig. 3. Both estimators give the same spectral shape, an inverse square power law $S_B(f) \propto f^{-2}$ spanning the entire accessible band from 52.6 Hz, set by the inverse record length, to the Nyquist frequency of 200 kHz. No knee is visible, so the Lorentzian form

$$S_B(f) = \frac{4\sigma_B^2\tau_c}{1+(2\pi f\tau_c)^2} \tag{20}$$

appropriate to Ornstein Uhlenbeck noise fits the data well but constrains only the product $\sigma_B^2\tau_c$, that is the high frequency amplitude $S_B f^2 = \sigma_B^2/\pi^2\tau_c$; the fit returns $\tau_c \approx 56$ ms with σ_B correspondingly inflated to 135 nT. Only a lower bound $\tau_c \gtrsim 19$ ms can honestly be quoted for the correlation time, and this is of no consequence for the coherence predictions below, which probe timescales far shorter than τ_c and therefore depend only on the well determined product. The direct autocorrelation estimator lies a factor of about 1.7 above the Welch estimate; this is spectral leakage, since the implicit rectangular data window has sidelobes falling as f^{-2}, parallel to the signal spectrum itself, so that low frequency power leaks uniformly across the band. The Welch estimate, whose Hann taper gives f^{-3} sidelobes, is used for all quantitative work.

B Coherence decay

Figure 4 shows the coherence decay for all five sequences and Table 1 collects the fitted parameters. The free induction decay gives

$$T_2^* = 71.0 \pm 0.2\ \mu\text{s}, \qquad p = 2.18 \pm 0.02, \tag{21}$$

Table 1: Extracted Coherence Times and Fit Parameters

Sequence	N_π	T_2 (μs)	p	T_2^{FF} (μs)
Ramsey	0	71 ± 0	2.18 ± 0.02	63
Hahn echo	1	1094 ± 1	2.93 ± 0.01	1060
CPMG-2	2	1738 ± 2	2.79 ± 0.01	1680
CPMG-4	4	2781 ± 15	2.52 ± 0.05	2666
CPMG-8	8	4407 ± 32	2.68 ± 0.08	4235

N_π is the number of refocusing pulses; T_2 and p are from the fit (19) to the simulated decay; T_2^{FF} is the parameter free prediction from the measured spectrum. Uncertainties are the fit standard errors.

in good agreement with the quasi static prediction $\sqrt{2}/\gamma\sigma_B = 84.5\ \mu$s from (9); the residual difference reflects the modest evolution of the field within 71 μs and the finite ensemble. The exponent close to two confirms that the noise is essentially frozen on this timescale.

A single refocusing pulse extends the coherence time by a factor of fifteen to 1094 μs, with the exponent rising to $p = 2.93 \pm 0.01$. Additional pulses give further gains, reaching $4407 \pm 32\ \mu$s for CPMG-8. The fitted exponents for the decoupled sequences lie between 2.5 and 2.9.

C Comparison with the spectral prediction

The dashed curves in Fig. 4 are computed from the fitted spectrum through (15) with no free parameters. They reproduce the simulated decays throughout, and the coherence times agree to within four to ten percent across all five sequences, as listed in Table 1. The agreement is the central consistency check of this work: the spectrum estimated from the traces contains, by itself, enough information to predict the outcome of the full density matrix simulation.

Figure 5 shows the decoupling gain against pulse number. A power law fit gives

$$T_2(N) \propto N^{0.67\pm0.02}, \tag{22}$$

indistinguishable from the exponent $2/3$ predicted for a spectrum falling as f^{-2} within the sequence passband.

VI Discussion

The results form a consistent picture in which a single spectral feature, the f^{-2} fall off, explains every observation.

The Ramsey decay is Gaussian because the spectrum is dominated by frequencies far below $1/T_2^*$. In this quasi static regime each spin experiences an essentially constant detuning drawn from a Gaussian of width σ_B, and the ensemble average of $e^{-i\gamma B\tau}$ is the characteristic function of that Gaussian, giving (9). The measured T_2^* of 71 μs against the predicted 84.5 μs is agreement at the level one should expect given that the quasi static formula neglects all dynamics.

The fifteenfold echo gain follows directly, since it is precisely the quasi static component that the echo cancels. What survives is the noise evolving on the timescale of the sequence itself, and for Lorentzian noise in the limit $\tau \ll \tau_c$ the leading term is $\chi(\tau) \propto \gamma^2\sigma_B^2\tau^3/\tau_c$, a cubic exponent.

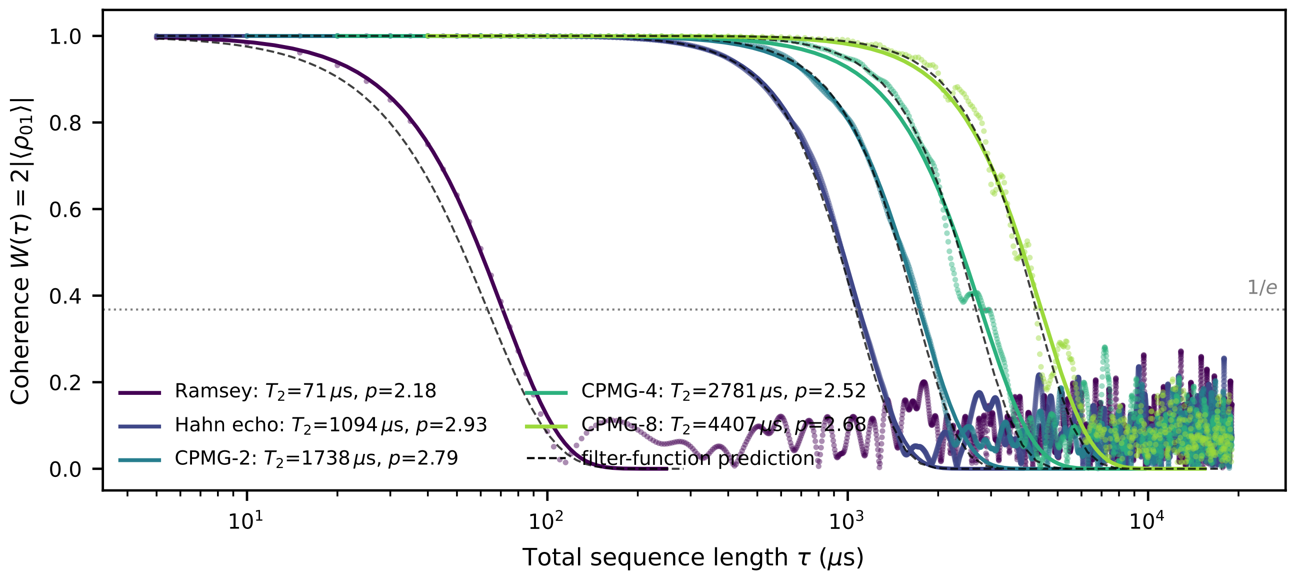

Figure 4: Simulated ensemble coherence against total sequence length for all five sequences. Points are the trace based density matrix simulation, solid lines are stretched exponential fits, and dashed black lines are parameter free predictions from the measured spectrum via the filter function integral (15). The scatter below $W \approx 0.1$ is the statistical floor of a one hundred spin ensemble.

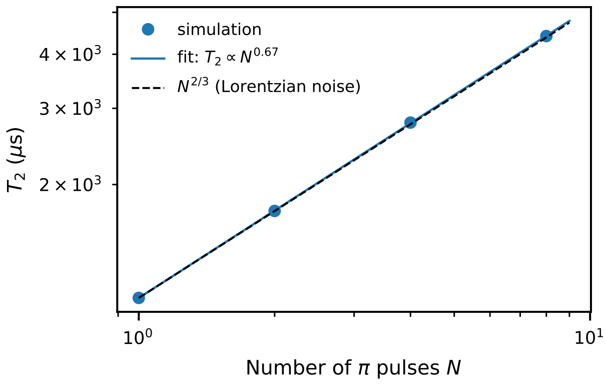

Figure 5: Scaling of the extracted coherence time with the number of refocusing pulses. The fitted exponent of 0.67 matches the two thirds law expected for noise with a Lorentzian spectrum in the regime $\tau \ll \tau_c$.

The fitted $p = 2.93$ for the Hahn echo is a clean confirmation. The slight softening towards $p \approx 2.5$ at larger N is expected: the CPMG filter passband is narrower and better approximates a single frequency, driving the decay back towards the Gaussian form characteristic of a monochromatic filter.

The $N^{2/3}$ scaling has the same origin. With the filter passband centred near $f \approx N/2\tau$ and the spectrum falling as f^{-2}, the decoherence integral scales as $\chi \propto \tau^3/N^2$, and setting $\chi = 1$ gives $T_2 \propto N^{2/3}$. The measured 0.67 leaves little room for doubt. A practical consequence is that decoupling here is a game of diminishing returns: each doubling of the pulse number buys only a factor $2^{2/3} \approx 1.6$ in coherence time, so the four hundredfold improvement from Ramsey to a hypothetical thousand pulse sequence would demand pulse fidelities far beyond what the ideal pulse assumption made here can justify.

Several limitations should be noted. All pulses are treated as instantaneous and perfect, so no error accumulates with pulse number; in practice finite pulse duration and rotation errors impose a ceiling on the achievable N, and CPMG is chosen over Carr Purcell precisely because it is tolerant of such errors. The ensemble of one hundred traces imposes a statistical floor of about 0.09 in coherence, restricting fits to the first decade of decay. The record length of 19 ms bounds the lowest resolvable frequency, so the correlation time is a lower bound rather than a measurement, and the longest CPMG-8 coherence times approach a non negligible fraction of the record. Finally, the model is one of pure dephasing: transverse field components that would drive population relaxation are absent by construction, so T_1 processes and the associated ceiling $T_2 \leq 2T_1$ do not appear.

VII Conclusion

An ensemble of one hundred measured magnetic field noise traces has been used to simulate spin dephasing by direct density matrix propagation under explicit rotation operators. The noise spectrum, estimated independently by Welch averaging and by transformation of the autocorrelation function, follows an inverse square power law across the accessible band. Free induction decay yields $T_2^* = 71$ μs with a Gaussian profile, a Hahn echo extends this to 1.09 ms, and CPMG with eight pulses reaches 4.41 ms, the gain scaling as $N^{2/3}$. Coherence times predicted from the measured spectrum through the filter function formalism agree with the simulation to within ten percent without adjustable parameters, demonstrating that the spectral characterisation and the trajectory level simulation are two consistent views of the same physics.

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